

Sinusoidal steady-state analysis

ELEC0447 - Analysis of electric power and energy systems

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Overview

1. Basics and conventions
2. Sinusoidal steady-state analysis

What will we learn today?

Mostly from Chapter 2 of Ned Mohan's book:

Mohan, Ned. *Electric power systems: a first course*. John Wiley & Sons, 2012.

- ▶ Recap of conventions
- ▶ Sinusoidal steady-state analysis

You will be able to do exercises 2.1, 2.2, 2.4, 2.5, 2.9, 2.11, 2.12, 2.14, 2.16, 2.17, 2.18, 2.19 and 2.20 from Ned Mohan's book.

Basics and conventions

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Power and energy

- ▶ Power measures the rate of use of energy.
- ▶ It is expressed in Watt [W]: $1 \text{ W} = 1 \text{ Joule/second}$.
- ▶ In an electric system, the instantaneous power is:

$$p(t) = u(t) \times i(t)$$

- ▶ $u(t)$ is the voltage in volt [V] (line integral of electric field between two points).
 - ▶ $i(t)$ is the current in amperes [A].
 - ▶ t is time.
- ▶ To measure energy in power systems, we use units ranging from a kWh (a microgrid) to a TWh (a country).
- ▶ Devices have power ratings ranging from W to GW (although we generally speak in VA for ratings).

Power in perspective: The cyclist vs. the toaster

Can an Olympic cyclist generate enough electrical power to toast a single slice of bread?

The Cyclist vs. The Toaster

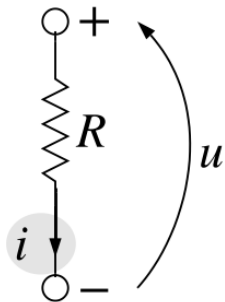
Robert Förstemann vs. 700W Toaster

Watch video: <https://www.youtube.com/watch?v=S405vo0CqAQ>

It takes an Olympic sprint cyclist generating nearly 700 W of mechanical power at maximum exertion for over a minute just to toast one single slice of bread!

Motor convention (or standard reference)

When using the **motor convention** to direct u w.r.t. i , $p(t)$ represents the power **consumed** by a device (here a resistor):



- ▶ The power consumed can be < 0 , $= 0$, or > 0 depending on the device.
- ▶ E.g., for a passive resistor we always have $p(t) \geq 0$.
- ▶ The **opposite** convention is the **generator convention** ($p(t)$ is the power produced).
- ▶ We will sometimes use a mix of both conventions, based on intuition, so that in general we have few negative numbers: **pay attention to the orientations!**

Magnetic fields, etc. I

Magnetic fields play a central role in modeling the behavior of power system equipment (lines, transformers, generators, etc.).

As this course will not focus on detailed internal device physics, but rather on a *system of devices*, magnetic effects will often be highly abstracted (e.g., as an inductance, or a turns ratio).

Let's recall that:

- ▶ A magnetic field is due to charges in motion or magnetized materials.

Magnetic fields, etc. II

- ▶ It is measured in A/m (for magnetic field strength H) or in Tesla [T] (for magnetic flux density $B = \mu H$, with μ_0 in vacuum/air).
- ▶ The magnetic flux (ϕ in Weber [Wb]) measures the magnetic field crossing a surface.
- ▶ A time-varying magnetic flux induces a voltage (Faraday-Lenz Law: $e(t) = -\frac{d\phi}{dt}$), which is the fundamental principle behind transformers and generators.

Sinusoidal steady-state analysis

The background of the slide features a white central area where the text is located. This central area is framed by two large teal-colored triangles that point towards each other from the bottom corners, meeting at a point just below the center of the slide.

Sinusoidal signals and phasor representation I

Unless otherwise specified, we will always work with sinusoidal signals and in steady state:

$$y(t) = \sqrt{2}Y \cos(\omega t + \phi_y)$$

- ▶ Y is the **rms** value of the signal.
- ▶ ϕ_y is its **phase angle**.
- ▶ $\omega = 2\pi f$ is its **angular frequency** in rad/s (f is the frequency in Hz).

Sinusoidal signals and phasor representation II

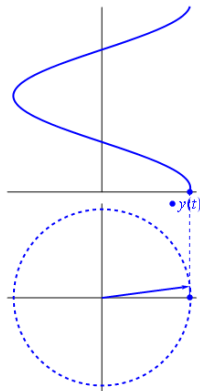
At a specific frequency $f = \frac{\omega}{2\pi}$, the signal can be represented as a **phasor**:

$$\bar{Y} = Y \angle \phi_y = Y e^{j\phi_y}$$

Phasors allow working in the frequency domain, converting differential equations into algebraic operations:

$$\mathcal{Re} \left\{ \sqrt{2} \bar{Y} e^{j\omega t} \right\} = y(t)$$

Phasor concept



- ▶ A phasor represents a sinusoidal waveform rotating counter-clockwise at angular speed ω .
- ▶ Its projection on the real axis corresponds to the instantaneous value $y(t)$:

$$y(t) = \mathcal{R}e\{\sqrt{2}Y e^{j(\omega t + \phi_y)}\}$$

- ▶ Factoring out $e^{j\omega t}$ allows us to perform additions, subtractions, and circuit laws purely using complex algebra.
- ▶ Reference:

<https://en.wikipedia.org/wiki/Phasor>

Impedance I

Let $u(t)$ and $i(t)$ be the voltage and current across a one-port, respectively, in sinusoidal steady state and with the motor convention:

- ▶ For a resistor: $u(t) = Ri(t) \implies \bar{U} = R\bar{I}$
- ▶ For an inductor: $u(t) = L\frac{di(t)}{dt} \implies \bar{U} = j\omega L\bar{I}$
- ▶ For a capacitor: $i(t) = C\frac{du(t)}{dt} \implies \bar{I} = j\omega C\bar{U} \implies \bar{U} = -j\frac{1}{\omega C}\bar{I}$

Impedance II

The impedance

The **impedance** Z , a complex quantity, generalizes Ohm's law:

$$Z = R + jX \quad [\Omega], \quad \bar{U} = Z\bar{I}$$

- ▶ Resistor: $Z = R$ ($X = 0$)
- ▶ Inductor: $Z = jX = j\omega L$ ($X = \omega L > 0$)
- ▶ Capacitor: $Z = jX = -j\frac{1}{\omega C}$ ($X = -\frac{1}{\omega C} < 0$)

Impedance, admittance, etc. I

The imaginary part of the impedance, X , is called **reactance**:

- ▶ $X > 0$: inductive reactance
- ▶ $X < 0$: capacitive reactance

The **admittance** Y is the inverse of the impedance, expressed in Siemens [S]:

$$Y = \frac{1}{Z} = G + jB$$

- ▶ G is the **conductance**

Impedance, admittance, etc. II

- B is the **susceptance**

Explicitly:

$$Y = \frac{1}{R + jX} = \frac{R - jX}{R^2 + X^2} \implies G = \frac{R}{R^2 + X^2}, \quad B = \frac{-X}{R^2 + X^2}$$

Complex calculus

For an impedance $Z = R + jX$:

$$|Z| = \sqrt{R^2 + X^2}, \quad \angle Z = \arctan\left(\frac{X}{R}\right)$$

$$Z = \frac{\bar{U}}{\bar{I}} = \frac{U \angle \phi_u}{I \angle \phi_i} = \frac{U}{I} \angle (\phi_u - \phi_i)$$

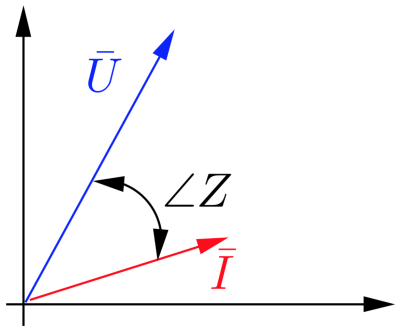
Animation

AC Circuit & Phasor Analysis Animation:

<https://www.youtube.com/watch?v=z07RZZW0wSQ>

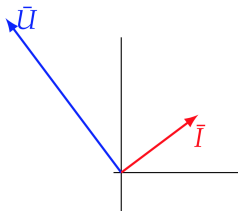
Phasor diagrams

Plotting the phasors in the complex plane provides immediate geometric insight:



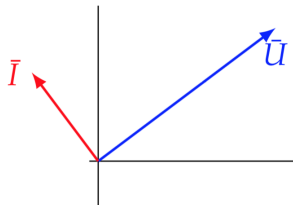
- ▶ One phasor (often the bus voltage $\bar{U} = U \angle 0^\circ$) is chosen as the reference along the positive real axis.
- ▶ The current phasor $\bar{I} = I \angle \phi_i$ is placed at angle ϕ_i .
- ▶ The phase difference is $\phi = \phi_u - \phi_i$.
- ▶ Angles are measured counter-clockwise (positive).

Inductive or capacitive? Which is which?



Inductive circuit ($X > 0$)

- ▶ Current $\bar{I}$ is **lagging** behind voltage $\bar{U}$
- ▶ $\phi_u > \phi_i \implies \phi = \phi_u - \phi_i > 0$
- ▶ Consumes reactive power ($Q > 0$)



Capacitive circuit ($X < 0$)

- ▶ Current $\bar{I}$ is **leading** ahead of $\bar{U}$
- ▶ $\phi_u < \phi_i \implies \phi = \phi_u - \phi_i < 0$
- ▶ Supplies reactive power ($Q < 0$ in motor conv.)

The notions of power I

The **complex power** is defined as:

$$S = \bar{U} \bar{I}^*$$

Let $\phi = \phi_u - \phi_i$, then:

$$S = (Ue^{j\phi_u})(Ie^{-j\phi_i}) = UIe^{j\phi} = P + jQ$$

- ▶ $P = UI \cos \phi$ is the **active power**, measured in Watt [W].
- ▶ $Q = UI \sin \phi$ is the **reactive power**, measured in volt-ampere reactive [var].

The notions of power II

- ▶ $\cos \phi$ is the **power factor** (pf).
- ▶ The **apparent power** is $|S| = UI = \sqrt{P^2 + Q^2}$, measured in volt-ampere [VA].

Reactive power is, in general, undesirable: it increases the current magnitude $I = |S|/U$ for a given active power P , resulting in higher line losses (RI^2) and larger voltage drops.

Useful formulas: leading and lagging circuits

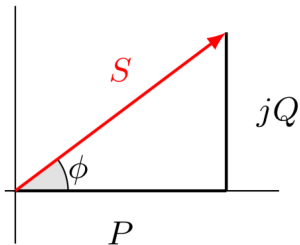
$$P = RI^2 = \frac{U_R^2}{R}, \quad Q = XI^2 = \frac{U_X^2}{X}$$
$$\tan \phi = \frac{Q}{P}, \quad \cos \phi = \frac{P}{|S|}$$

Leading vs. Lagging

The value of the power factor $\cos \phi \in [0, 1]$ alone does not indicate the sign of ϕ :

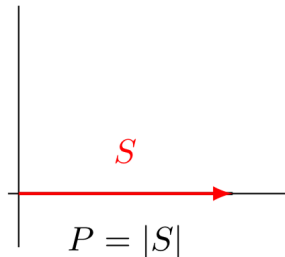
- ▶ In an **inductive** system, $u(t)$ precedes $i(t)$, $i(t)$ is **lagging**, thus $Q > 0$ (motor convention).
- ▶ In a **capacitive** system, the opposite holds: $i(t)$ is **leading**, thus $Q < 0$.

Power factor compensation



Uncompensated load draws active power P and reactive power $Q_L > 0$. Total line current:

$$I_1 = \frac{\sqrt{P^2 + Q_L^2}}{U}.$$



Connecting a parallel capacitor provides $Q_C = -\omega CU^2 < 0$. Total reactive power is canceled: $Q_{net} \approx 0$. Total current drops to $I_2 \approx \frac{P}{U} < I_1$!

Power factor compensation: Example

Problem statement

A 120 V (rms) voltage source at 60 Hz feeds an R-L load:

$$S_{load} = 1858.4 + j1031.4 \text{ VA}$$

- ▶ Active power: $P = 1858.4 \text{ W}$
- ▶ Reactive power: $Q = 1031.4 \text{ var}$ (inductive, lagging)
- ▶ Apparent power: $|S| = \sqrt{P^2 + Q^2} = 2125 \text{ VA}$
- ▶ Initial power factor: $\cos \phi = \frac{P}{|S|} = \frac{1858.4}{2125} \approx 0.875$ lagging
- ▶ Initial line current: $I = \frac{|S|}{U} = \frac{2125}{120} \approx 17.7 \text{ A}$

Power factor compensation: Example (cont'd)

Goal: Compensate to unity power factor ($\cos \phi_{new} = 1.0$, $Q_{new} = 0$).

The parallel capacitor must supply reactive power equal and opposite to Q_{load} :

$$Q_C = -Q_{load} = -1031.4 \text{ var} = -\omega C U^2$$

Solving for C :

$$C = \frac{Q_{load}}{\omega U^2} = \frac{1031.4}{2\pi \times 60 \times 120^2} \approx 189.9 \mu\text{F}$$

Result on line current:

$$I_{new} = \frac{P}{U} = \frac{1858.4}{120} \approx 15.49 \text{ A}$$

This represents a **12.5% reduction in line current** with zero change in delivered active power P !

Summary

- ▶ **Conventions:** Motor convention defines consumed power $p(t) = u(t)i(t)$; generator convention defines produced power.
- ▶ **Phasors:** Represent sinusoidal quantities $y(t) = \sqrt{2}Y \cos(\omega t + \phi_y)$ as algebraic complex numbers $\bar{Y} = Y e^{j\phi_y}$.
- ▶ **Impedance & Admittance:** $Z = R + jX$ and $Y = G + jB = 1/Z$.
- ▶ **Complex Power:** $S = \bar{U}\bar{I}^* = P + jQ = |S|e^{j\phi}$.
 - ▶ $P = UI \cos \phi$ (active power, in Watt [W])
 - ▶ $Q = UI \sin \phi$ (reactive power, in var [var])
 - ▶ $|S| = UI$ (apparent power, in Volt-Ampere [VA])
- ▶ **Power Factor:** $\cos \phi = P/|S|$; lagging for inductive loads, leading for capacitive loads.
- ▶ **Power Factor Correction:** Shunt capacitors supply reactive power locally, reducing current flows, thermal losses, and voltage drops across the network.